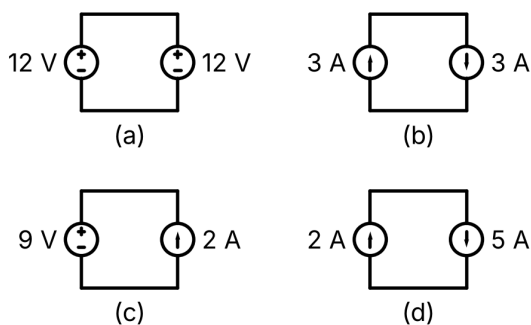


Practice sheet: Chapter 2, Circuit Elements

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz at the start of week 3 is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Both must be right.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Signs:** in KCL, count currents leaving a node as positive. In KVL, walk clockwise and count voltage drops (+ to −) as positive. $p > 0$ means an element absorbs power.

S1. Which connections are allowed? (two-tier) Each panel joins two ideal sources.



Which connection is **not** permitted?

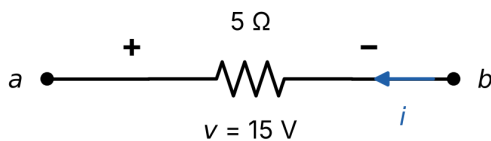
- a) (a) b) (b)
c) (c) d) (d)

Reason:

- Two ideal current sources in one loop must carry the same current, so 2 A and 5 A conflict.
- An ideal voltage source can never be connected in parallel with an ideal current source.
- Two ideal voltage sources can never be connected in parallel.
- The 5 A source would overload the 2 A source.

S2. Reading a dependent source. A source drawn as a diamond containing an arrow is labelled $0.4v_x$, where v_x is the voltage across a resistor elsewhere in the circuit. Which description is correct?

- A current-controlled voltage source; 0.4 is in V/A
- A voltage-controlled current source; 0.4 is in A/V
- A voltage-controlled voltage source; 0.4 has no unit
- An independent current source of 0.4 A

S3. Ohm's law and the reference arrow (two-tier).Find the current i .

- a) 3 A b) -3 A
c) 75 A d) -75 A

Reason:

- i) The arrow enters the + terminal, so $v = iR$.
ii) The arrow enters the - terminal, so $v = -iR$.
iii) Current flows from + to -, so i is positive.
iv) $i = v/R$ for every resistor, whatever the arrow.

Follow-up: through the resistor, does positive charge actually move from a to b or from b to a?

S4. Power in that resistor (two-tier). For the resistor in S3, the power is:

- a) 45 W absorbed b) 45 W delivered
c) 75 W absorbed d) 0 W

Reason:

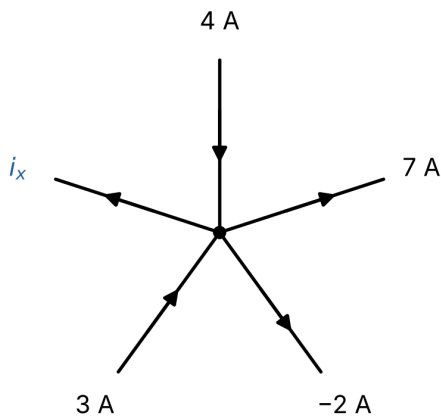
- i) $p = vi = (15)(-3) = -45$ W, so it delivers.
ii) The arrow enters the - terminal, so $p = -vi = 45$ W; a resistor always absorbs, $p = i^2R$.
iii) i is negative, so no power flows.
iv) $p = v^2/R$.

S5. Resistor ratings. A 1 kΩ resistor is rated ¼ W. What is the largest voltage you can put across it without exceeding its rating?

- a) 250 V b) 15.8 V
c) 0.25 V d) 4 V

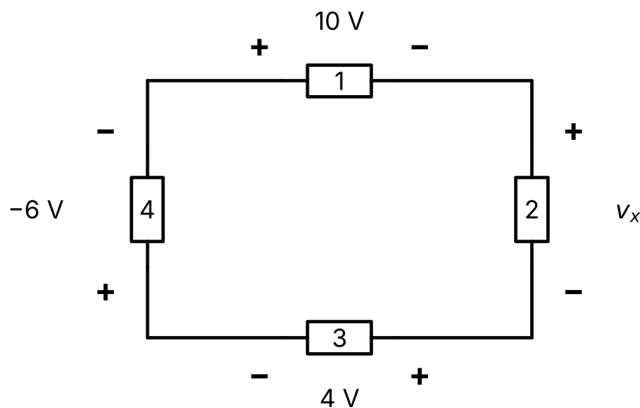
S6. A heater on your generator line (predict first). An electric heater is rated 2000 W at 220 V.

- a) Find its resistance and the current it draws.
b) Can it run on a 5 A generator subscription?
c) *Predict first:* if the line voltage sags to 200 V (9% lower), does the heater's power fall by less than 9%, about 9% or more than 9%? Then compute it, treating R as constant.

S7. KCL at a node (predict first).

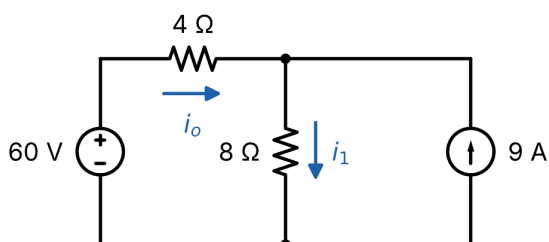
Predict first: which currents count as leaving the node? Then find i_x .

- | | |
|--------|---------|
| a) 2 A | b) -2 A |
| c) 6 A | d) -6 A |

S8. KVL around a loop.

Predict first: on a clockwise walk, which elements do you cross from + to -? Then find v_x .

- | | |
|---------|---------|
| a) -8 V | b) 8 V |
| c) 0 V | d) 20 V |

S9. Two unknown currents.

- Write a KCL equation at the top node and a KVL equation around the left loop.
- Find i_o and i_1 .
- Find the power of each source, say whether it absorbs or delivers, and check the power balance.

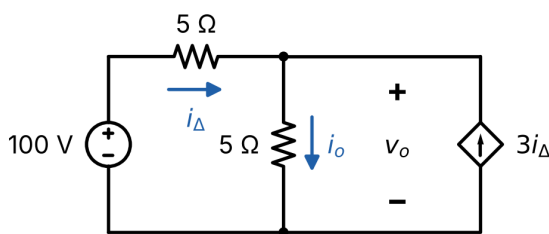
S10. Which loop gives a useful KVL equation? (two-tier) In the S9 circuit, which closed path gives a KVL equation without adding a new unknown?

- a) The loop through the 60 V source, the 4 Ω and the 8 Ω resistors
- b) The loop through the 8 Ω resistor and the 9 A source
- c) The outer loop through the 60 V source, the 4 Ω resistor and the 9 A source
- d) All three are equally useful

Reason:

- i) The voltage across an ideal current source is unknown; the rest of the circuit sets it.
- ii) The voltage across a current source is always zero.
- iii) KVL does not apply to a loop that contains a current source.
- iv) An outer loop never gives an independent equation.

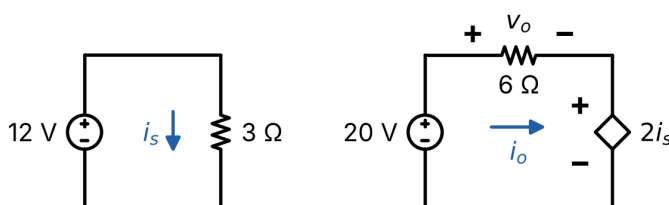
S11. A current-controlled current source.



Find i_Δ and v_o , and the power of the dependent source.

- a) $v_o = 80$ V
- b) $v_o = 50$ V
- c) $v_o = 200$ V
- d) $v_o = 100$ V

S12. A current-controlled voltage source (two-tier).



The dependent source:

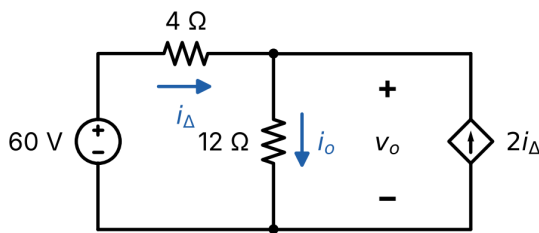
- a) absorbs 16 W
- b) delivers 16 W
- c) absorbs 8 W
- d) delivers 24 W

Reason:

- i) i_o enters its + terminal, so $p = v_i > 0$.
- ii) Dependent sources always deliver power.
- iii) Its voltage is positive, so it delivers.
- iv) It is controlled by i_s , so its power is zero.

Follow-up: find v_o .

S13. Spot the error. The worked solution below finds v_o . It contains the kind of mistake AI chat tools often make with dependent sources.



1. KVL clockwise around the left loop, drops positive: $-60 + 4i_{\Delta} + 12i_o = 0$.
 2. KCL at the top node, currents leaving: $-i_{\Delta} + i_o + 2i_{\Delta} = 0$, so $i_o = -i_{\Delta}$.
 3. Substitute: $-60 + 4i_{\Delta} - 12i_{\Delta} = 0$, so $i_{\Delta} = -7.5$ A.
 4. Then $i_o = 7.5$ A and $v_o = 12i_o = 90$ V.
 5. v_o is positive, so the answer is correct.
- a) Which is the first wrong line?
 - b) Correct it.
 - c) Finish the solution and check it with a power balance.
 - d) Line 5 claims the answer is correct. Why is that not a check?

S14. Work backwards. In the S9 circuit, replace the 9 A source with a source of unknown current I , arrow still up. What value of I makes the current in the 60 V source zero?

- a) 7.5 A
- b) 15 A
- c) 5 A
- d) 0 A

S15. Modelling a car battery from measurements. A car battery is tested at three load currents:

Current drawn (A)	0	100	200
Terminal voltage (V)	12.6	11.6	10.6

Model it as an ideal voltage source in series with a resistance, as in Nilsson Example 2.9. What terminal voltage do you expect when the starter motor draws 250 A?

- a) 10.1 V
- b) 12.6 V
- c) 9.1 V
- d) 11.35 V

Follow-up: why do the headlights dim while the engine is cranking?

S16. Two heaters: series or parallel? (two-tier) Two identical heaters are each rated 1100 W at 220 V. You connect them in series across 220 V. *Predict first:* more heat than one heater alone, or less? Then find the total power.

- a) 2200 W
- b) 1100 W
- c) 550 W
- d) 275 W

Reason:

- i) Each heater keeps $R = 44 \Omega$; in series the current is $220/88 = 2.5$ A, so $P = 220 \times 2.5 = 550$ W.
- ii) Each heater still gets 220 V, so the powers add.
- iii) Two heaters always give twice the heat of one.
- iv) Each heater gets half the voltage, so it gives half its rated power.